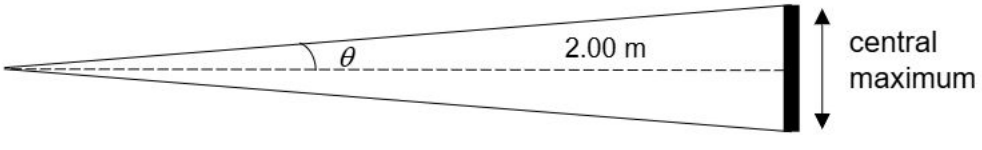
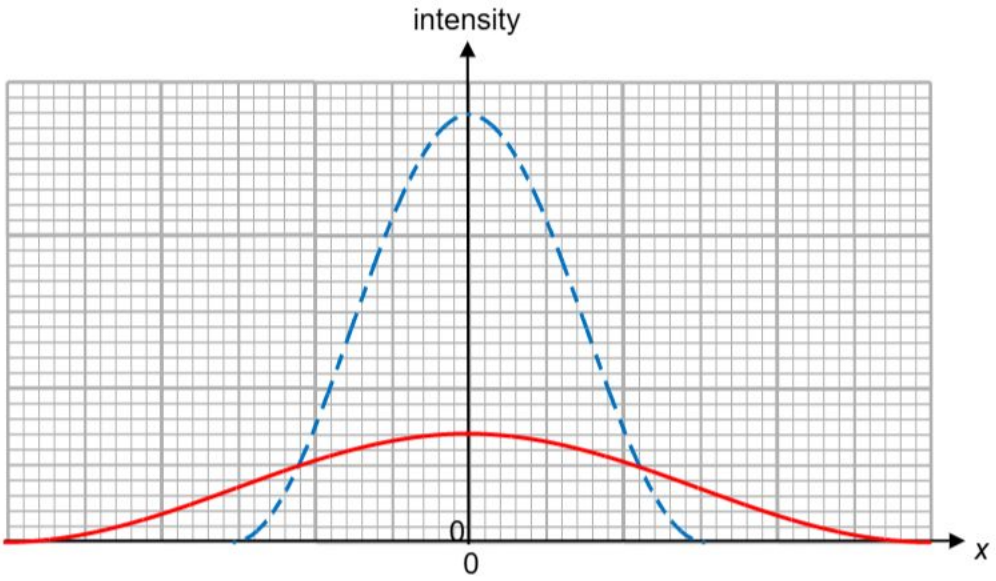
Question 10		
(a)	Coherent light waves are electromagnetic waves that have a **constant phase difference** between them.	**B1**
(b)(i)	Bright fringes are formed by the **constructive interference** of the coherent waves emerging from the two slits arriving on the screen **in phase**. [or with a constant phase difference of m x 2π rad where m: order (m= 0,1, 2 3…).] Dark fringes are formed by the **destructive interference** of the coherent waves arriving on the screen **out of phase.** [or with a constant phase difference of $(m-\frac{1}{2})\pi$ rad.] Note to marker: will only accept phase difference rather than path difference as phase difference is the fundamental condition for the outcome of the interference	**B1** **B1**
(b)(ii)	Using the fringe separation for double slit interference: fringe separation, y = $\frac{\lambda L}{d}$ = $\frac{(633 \times 10^{-9})(2.00)}{0.250 \times 10^{-3}}$ = 5.064×10^{-3} m Distance between the central bright fringe and the 2nd order dark fringe = 1.5y = 1.5 x 5.064 x 10^{-3} = 7.596 x 10^{-3} m = 7.60 mm	**C1** **A1**
(b)(iii)	The double-slit interference fringes are modulated by the single-slit diffraction envelope resulted from the appreciable width of each slit. As the single-slit diffraction envelope has its greatest intensity at the centre, the maximum observed at O is of a greater intensity than any other observable maximum.	**B1** **B1**
(c)(i)	The angle θ of first minimum for the single slit is given by: sin $\theta = \frac{\lambda}{b}$ sin $\theta = \frac{633\times10^{-9}}{0.080\times10^{-3}}$ = 0.0079125 For small angle θ, $\sin\theta \approx \tan\theta$  ∴width of the central maximum = 2 x 2.00 (tan θ) = 2 x 2.00 x 0.0079125 = 0.03165 m = 31.7 mm	**B1** **B1** **A0**

	Note to marker: Students who do not use small angle approximation will have the same value.	
(c)(ii)	 Marking point: • Peak intensity is smaller. (1/4 of original intensity not required) • Width doubles	**A1** **A1**
(c)(iii)	Using the Rayleigh criterion for the resolving power of a single aperture $\theta_m \approx \frac{\lambda}{b}$ Using small angle approximation for the angle subtended by the two sources at the aperture, $\theta_m \text{ (in rad)} \approx \frac{s}{D}$ $\Rightarrow \frac{s}{D} = \frac{\lambda}{b}$ $\therefore D = \frac{s \times b}{\lambda} = \frac{(5.0 \times 10^{-3})(0.080 \times 10^{-3})}{633 \times 10^{-9}}$ $= 0.632$ $= 0.63$ m	**B1** **C1** **A1**
(d)(i)	Maximum angle of diffraction $\theta = \tan^{-1}\left(\frac{3.00}{2.00}\right) = 56.31^o$ From the grating equation $d \sin\theta = n\lambda$, $n = \frac{\sin\theta}{N\lambda}$ where N is the number of slits per metre $n = \frac{\sin 56.31^o}{(500 \times 10^3)(633 \times 10^{-9})} = 2.6$ Hence maximum observable order = 2 Maximum number of observable bright fringes = 2(2) + 1 = 5	**B1** **B1** **C1** **A1**

(d)(ii)	The **first order maximum** shows a **continuous spectrum of colours** (with red furthest from the central maximum and violet closest to the central maximum) while **the central maximum shows a** (narrow) **white line/spot.** From the diffraction equation $d \sin\theta = n\lambda$. For $n = 0$, all the wavelengths overlap, producing the white line/spot in the central maximum. In the first order maximum ($n = 1$), **different wavelengths deviate with different angles** θ with the longer wavelength deviating with the larger angle hence producing the spread of colours in the continuous spectrum.	**B1** **B1**